

Ahmad D. Suleiman

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Research Profile

Computing education and empirical software engineering researcher developing adoption-ready project-based learning and responsible applications of large language models. Co-developed 14 NSF-funded computing projects adopted at seven U.S. and Canadian universities and impacting thousands of students. Author or co-author of nine peer-reviewed journal and conference papers, including three accepted 2026 papers.

Research Interests: Computing Education, Project-Based Learning, Empirical Software Engineering, Applied Large Language Models, AI-Assisted Computing Education

Education

Rochester Institute of Technology (RIT) <i>Ph.D. in Computing and Information Science</i>	Rochester, NY, USA <i>Expected Dec. 2026</i>
Clarkson University <i>M.S. in Electrical and Computer Engineering, Software Engineering GPA: 4.0/4.0</i>	Potsdam, NY, USA <i>May 2024</i>
Federal University of Technology, Minna (FUT Minna) <i>B.Tech. in Information Technology First Class Honors GPA: 4.58/5.0</i>	Minna, Nigeria <i>Nov. 2019</i>

Publications

Peer-Reviewed Journal Article

- J1. Suleiman, A. D.,** Hou, D., Liu, Y., DeWaters, J., Shepherd, D. C., & De Souza, J. G. "Systematic Literature Review on Project-Based Learning in Computing Education." *ACM Transactions on Computing Education*, 25(4), 2025. [doi:10.1145/3743684](https://doi.org/10.1145/3743684)

Peer-Reviewed Conference Papers

- C8. Suleiman, A. D.,** Hou, D., & Raida, M. N. "LLM-Generated Design Problems for Assessing Higher-Order Thinking in Project-Based Learning." *ACM Virtual Global Computing Education Conference V.1*, 2026. Accepted. [doi:10.1145/3795867.3831014](https://doi.org/10.1145/3795867.3831014) | [arXiv:2607.11032](https://arxiv.org/abs/2607.11032)
- C7. Suleiman, A. D.,** DeWaters, J., Shepherd, D., Korkmaz, T., Hussain, F., Liu, Y., & Hou, D. "Adoption-Ready Project-Based Learning for Computing Education: The FORAP Framework and a Multi-Scale Project Portfolio." *IEEE Frontiers in Education Conference*, 2026. Accepted. [arXiv:2607.11129](https://arxiv.org/abs/2607.11129)
- C6. Chen, Z., Ahir, L. J., Suleiman, A. D.,** Yao, K., Tang, Y., Shang, W., & Hou, D. "Do Privacy Policies Match with the Logs? An Empirical Study of Privacy Disclosure in Android Application Logs." *International Conference on Evaluation and Assessment in Software Engineering*, 2026. Accepted. [arXiv:2604.18552](https://arxiv.org/abs/2604.18552)
- C5. Suleiman, A. D.,** Tang, Y., & Hou, D. "Factors Impacting Faculty Adoption of Project-Based Learning in Computing Education: A Survey." *IEEE Frontiers in Education Conference*, 2025. [doi:10.1109/FIE63693.2025.11328475](https://doi.org/10.1109/FIE63693.2025.11328475)
- C4. Suleiman, A. D.,** DeWaters, J., Hou, D., Liu, Y., & Shepherd, D. "An Exploratory Comparative Study on the Impacts of Technical Support on Student Successes in Computing Project-Based Learning." *IEEE Frontiers in Education Conference*, 2024. [doi:10.1109/FIE61694.2024.10893440](https://doi.org/10.1109/FIE61694.2024.10893440)
- C3. Suleiman, A. D.,** Shepherd, D., DeWaters, J., Liu, Y., & Hou, D. "Providing Technical Support to Sustain Student Motivation and Engagement in Software Engineering Project-Based Learning." *IEEE Frontiers in Education Conference*, 2024. [doi:10.1109/FIE61694.2024.10893063](https://doi.org/10.1109/FIE61694.2024.10893063)
- C2. Suleiman, A. D.,** Hou, D., Liu, Y., DeWaters, J., Small, M. M., De Souza, J. G., & Shepherd, D. "Mapping Learning Objectives of Project-Based Undergraduate Software Engineering Courses to CC2020 Competency Model." *IEEE Frontiers in Education Conference*, 2023. [doi:10.1109/FIE58773.2023.10342967](https://doi.org/10.1109/FIE58773.2023.10342967)
- C1. De Souza, J. G., Flavell, M., Suleiman, A. D.,** Shepherd, D., DeWaters, J., Small, M. M., Liu, Y., & Hou, D. "The Importance of Project-Scale Scaffolding for Retention and Experience in Computing Courses." *IEEE Frontiers in Education Conference*, 2023. [doi:10.1109/FIE58773.2023.10343225](https://doi.org/10.1109/FIE58773.2023.10343225)

Manuscripts Under Review

- P3.** Chen, Z., Parmar, R. V., **Suleiman, A.**, Zhu, D., Yao, K., Tang, Y., Shang, W., & Hou, D. “Revisiting Android Fault Diagnosis Through the Lens of Third-Party Library Logs: An Empirical Study.” *ACM/IEEE International Symposium on Empirical Software Engineering and Measurement*, 2026. Under review.
- P2.** Tang, Y., Chen, Z., **Suleiman, A.**, Hou, D., & Shang, W. “Integrating Continuous Integration into Software Engineering Education Using GitHub Actions.” *IEEE Software*, 2025. Under review.
- P1.** Chen, Z., Nava-Camal, V., **Suleiman, A.**, Tang, Y., Hou, D., & Shang, W. “An Empirical Study on the Capability of LLMs in Decomposing Bug Reports.” *Science of Computer Programming*, 2026. Under review. [arXiv:2504.20911](https://arxiv.org/abs/2504.20911)

Research Experience

Research Assistant

Jan. 2022 – Present

Rochester Institute of Technology & Clarkson University

Rochester, NY

- Co-developed and deployed 14 NSF-funded project-based learning projects in software engineering, artificial intelligence, and cybersecurity, adopted at seven U.S. and Canadian universities and impacting thousands of students.
- Developed FORAP, the expert-validated Framework for Organizing Reusable and Adaptable PjBL Projects, to support project adoption and adaptation across computing education contexts.
- Led and contributed to empirical studies of project design, student technical support, faculty adoption, higher-order thinking assessment, learner profiles, and student interaction data.
- Built retrieval-augmented assistants, fine-tuned models, and agentic workflows with API integrations for project scaffolding, student support, and rubric generation.
- Contributed to empirical software engineering studies of Android application logs, privacy disclosure, third-party library logs, continuous integration in software engineering education, and LLM-based bug report decomposition.
- Configured and maintained a 20-node Red Hat OpenHPC cluster with 640 CPU cores, Slurm, OpenMPI, and Prometheus/Grafana monitoring for research simulations.

Teaching and Mentoring

- **Teaching Assistant, NSF Project-Based Learning Initiatives** (2022–Present) – Supported software engineering, artificial intelligence, and cybersecurity courses across multiple institutions. Assisted students with project implementation, led laboratory sessions, and supported faculty integration of project modules.
- **Undergraduate Research Mentor, NSF Research Experiences for Undergraduates** (2024) – Mentored undergraduate researchers investigating the capability of large language models to attempt student software engineering projects.
- **Teaching Assistant, NSF CyberTraining on High-Performance Computing** (2024) – Assisted participants with hands-on high-performance computing concepts, tools, and exercises.
- **Mentor, Michael Taiwo Graduate Scholarships** (2023–Present) – Mentor undergraduate and early-career scholars on graduate applications, research opportunities, statement writing, research planning, and graduate funding.

Professional Software Engineering Experience

Software Engineer

Apr. 2021 – Dec. 2021

Legalfit LLC

Overland Park, KS

- Developed and optimized content-management features for law firm websites, helping pages meet Google Core Web Vitals thresholds for LCP, INP, and CLS. *Django, Vue.js, PostgreSQL, AWS.*
- Implemented six responsive frontend templates from Adobe XD designs. *Vue.js, Jinja, HTML, CSS, JavaScript.*

Software Engineer

Jun. 2020 – Apr. 2021

TTF Ltd.

Lagos, Nigeria

- Built a public registration platform for Lagos State that scaled to more than 200,000 daily users. *Laravel, PHP, MySQL.*
- Built an end-to-end HR management system adopted by Sterling Bank. *Java, Spring, MySQL.*
- Designed an online ordering system for Nuli Lounge with administrative order management. *CodeIgniter, MySQL.*

Academic Service

- **Program Committee Member and Reviewer**, ACM Virtual Global Computing Education Conference, 2026 – Reviewed three papers.
- **Program Committee Member and Reviewer**, ACM Technical Symposium on Computer Science Education, 2026 – Reviewed three papers.
- **Peer Reviewer**, IEEE Frontiers in Education Conference, 2026 – Reviewed three papers.
- **Peer Reviewer**, IEEE Frontiers in Education Conference, 2025 – Reviewed seven papers.
- **Peer Reviewer**, IEEE Frontiers in Education Conference, 2024 – Reviewed three papers.

Awards and Scholarships

- **Graduate Research Assistantship**, Rochester Institute of Technology and Clarkson University, 2022–2026 – Full doctoral funding in Computing and Information Science.
- **Best Presentation Award**, Clarkson University Research and Project Showcase, 2023.
- **Michael Taiwo Scholarship**, 2020 – Merit-based scholarship supporting graduate applications through financial support and mentorship.
- **Shell SPDC Scholarship**, 2014–2019 – Nigerian national undergraduate scholarship for high-achieving STEM students.

Technical Skills

Languages: Python, Java, C++, PHP, JavaScript, Bash, SQL, HTML/CSS

AI/ML & LLMs: PyTorch, Hugging Face Transformers, LangChain, LangGraph, RAG, PEFT, Prompt Engineering, LLM Agents, Embeddings

Backend, APIs & Web: Flask, Django, Spring Boot, Laravel, REST APIs, React, Vue, Tailwind, Jinja2

Cloud, Data & Tools: AWS, Google Cloud Platform, Docker, Git, CI/CD, Data Pipelines, CUDA, OpenMPI, Slurm, Red Hat OpenHPC, PostgreSQL, MySQL

Professional Memberships

- **Association for Computing Machinery (ACM)** – Professional member since 2024
- **ACM Special Interest Group on Computer Science Education (SIGCSE)** – Member since 2023
- **American Society for Engineering Education (ASEE)** – Member in DSAI and CIT divisions 2025
- **Institute of Electrical and Electronics Engineers (IEEE)** – Member since 2023
- **IEEE Young Professionals** – Active member since 2023